



Route Optimization Software Methodology



GFATMTB(RSSH)GC7/FRQ-01/2025-2026



Proposed Methodology

1. Introduction

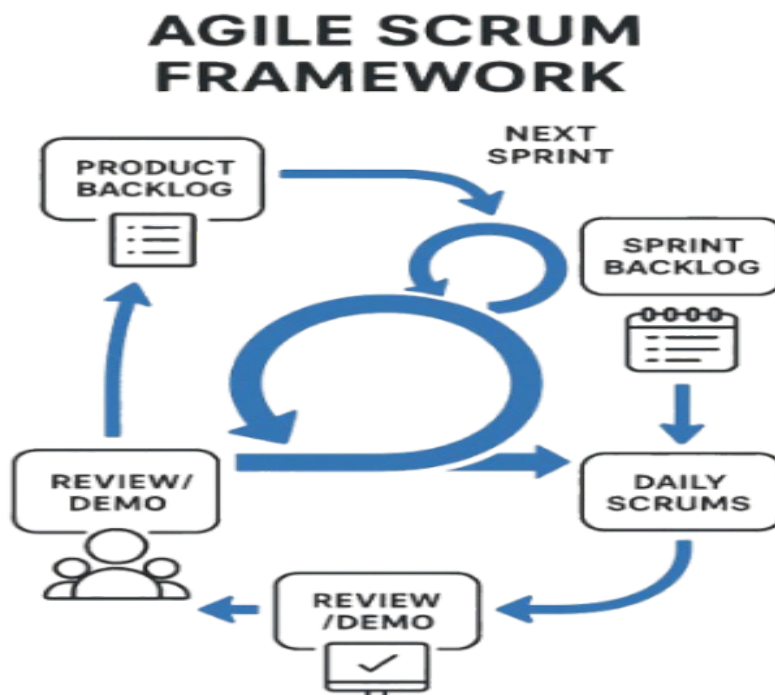
The implementation of the Route/Network Optimization Software for KEMSA will follow an **Agile–Scrum methodology**, selected for its flexibility, transparency, and alignment with continuous stakeholder engagement. Given that KEMSA’s logistics operations evolve dynamically—with frequent updates to routes, health facility demand, and distribution schedules—Agile’s iterative model ensures that the solution can adapt in real time while maintaining compliance with technical and operational specifications outlined in the RFQ.

2. Methodological Framework

Agile–Scrum emphasizes incremental delivery, stakeholder collaboration, and continuous improvement. Each sprint produces working software components (e.g., route engine, dashboard, reporting). This approach minimizes risk and accelerates value delivery.

Agile–Scrum Lifecycle

Figure 1: Agile–Scrum Lifecycle for KEMSA Software Implementation.



3. Project Phases and Deliverables

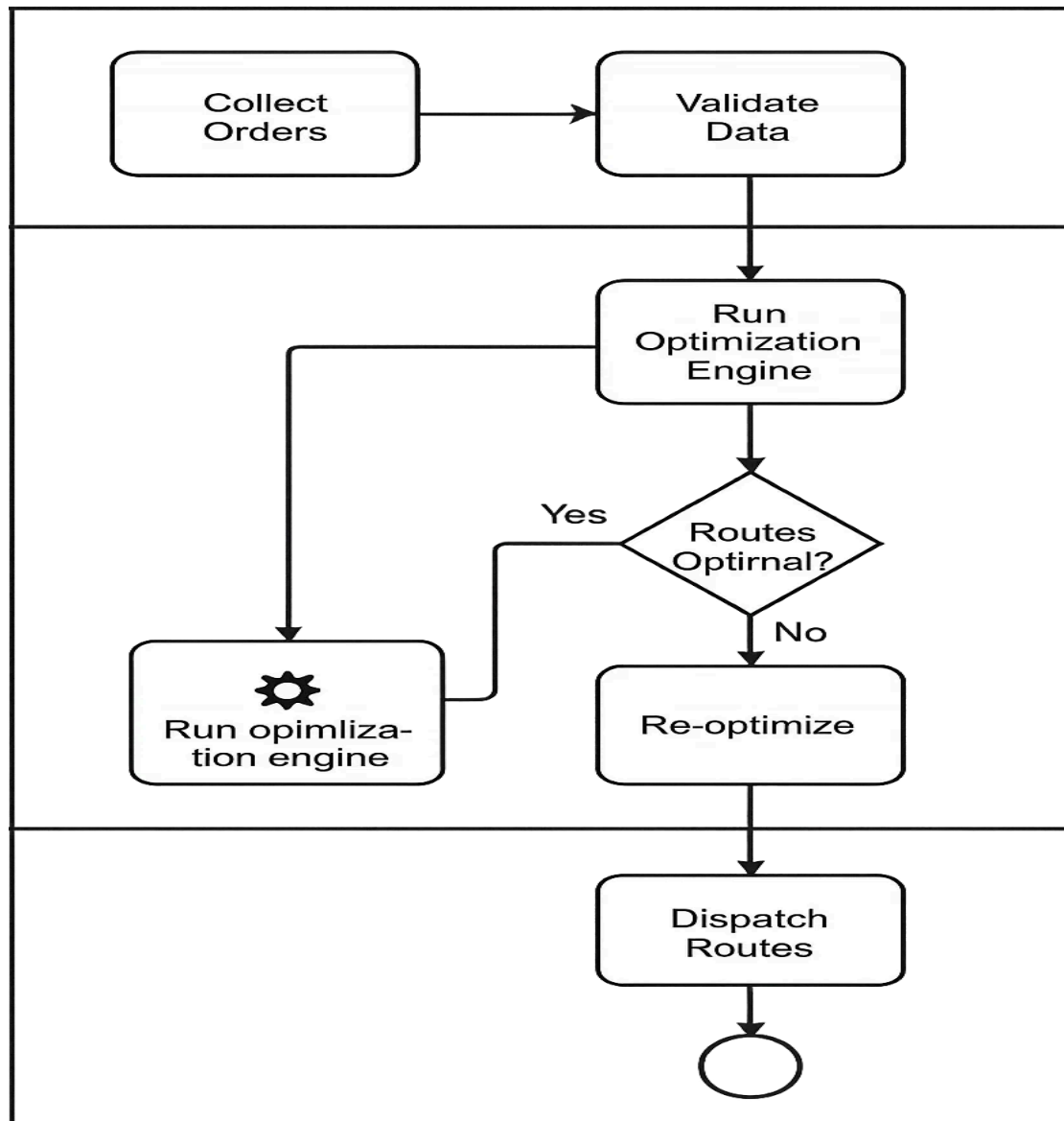
The project will progress through structured, overlapping phases—each corresponding to a set of sprints.

Phase	Duration	Key Activities	Outputs/Deliverables	KEMSA Involvement
Initiation & Planning	Week 1–2	Requirement analysis, stakeholder onboarding, risk assessment	Approved project plan, backlog, and resource matrix	Approve scope and kickoff
System & Architecture Design	Week 3–4	Design of software modules, data models, deployment architecture	System architecture diagrams, BPMN workflow	Review architecture
Development Sprints	Week 5–9	3 sprints (data module, optimization engine, UI/reporting)	Working increments, sprint reviews, test reports	Participate in demos
Deployment & Integration	Week 10	Installation on Ubuntu 24.04 server, environment configuration	Deployed solution, UAT sign-off	Verify performance
Training & Support	Week 11–12	End-user training, knowledge transfer, support setup	Training manuals, handover documentation	Attend sessions

4. Business Process Model and Notation Workflow

To visualize the operational flow, the process can be modeled in BPMN showing major roles (Planner, System, Driver).

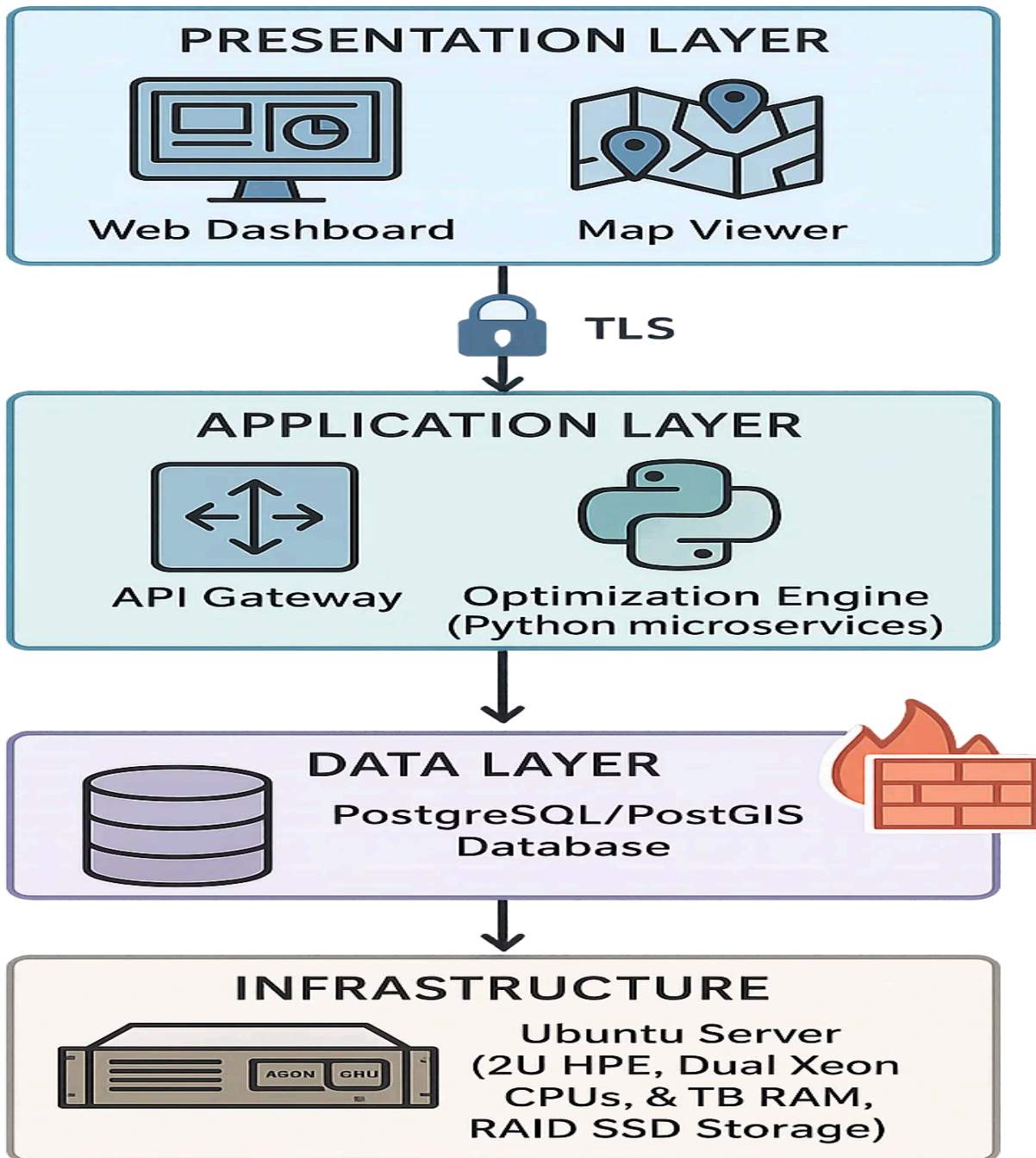
Figure 2: BPMN Process for Route Optimization Workflow.



5. System Architecture and Infrastructure

The Route Optimization System will adopt a three-tier architecture (Frontend, Backend, Database), deployed on KEMSA's dual-socket 2U rack server running Ubuntu 24.04 LTS, as specified in the RFQ.

Figure 3: System Architecture Diagram.



Layer	Components	Function	Tools/Tech
Presentation	Web/Mobile UI	Visualization of routes, reports, KPIs	ReactJS, Mapbox, HTML5
Application	Route Engine & API	Compute optimal paths, manage data exchange	Python, Flask, RESTful APIs
Data	Database & GIS	Store coordinates, demand, schedules	PostgreSQL, PostGIS
Infrastructure	Hardware/Network	High-availability hosting	Ubuntu 24.04, RAID NVMe, 100 GbE NICs

6. Agile Sprint Plan

Each sprint will deliver a functional increment tested and reviewed by KEMSA stakeholders.

Sprint	Duration	Objectives	Deliverables
Sprint 1	2 weeks	Set up environment, data ingestion pipeline	Data import module, connectivity test
Sprint 2	2 weeks	Develop core optimization algorithms	Working route engine, test dataset
Sprint 3	2 weeks	Build dashboards, analytics, reports	Interactive dashboard prototype
Sprint 4	2 weeks	Integration, security hardening, UAT	Stable release candidate
Sprint 5	2 weeks	Deployment, training, documentation	Production version, training materials

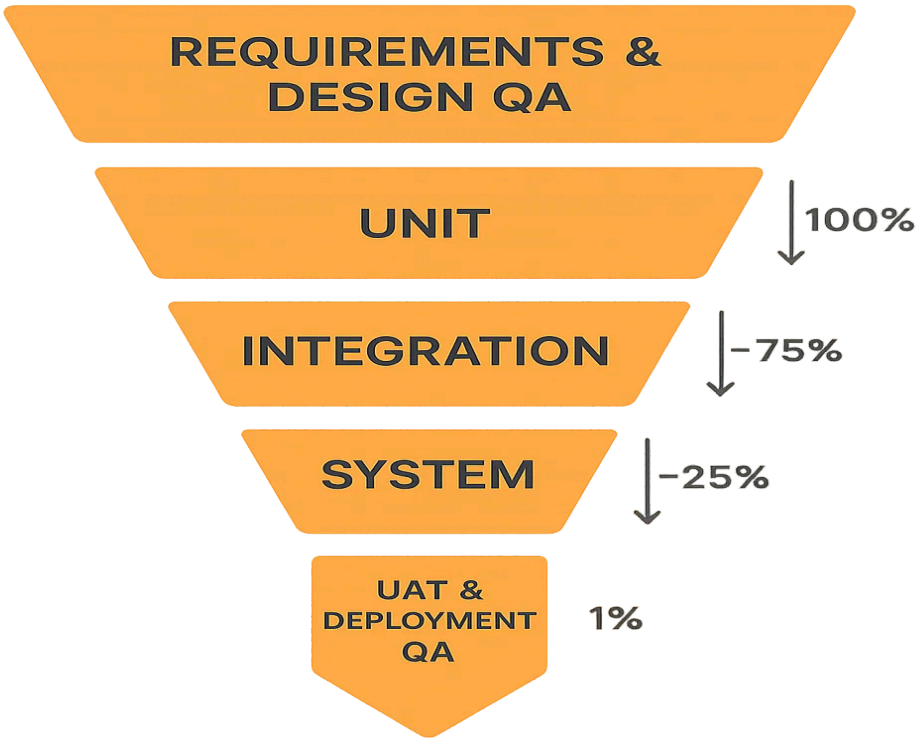
7. Quality Assurance (QA) and Testing

Quality management is embedded throughout. Testing occurs at multiple levels:

- **Unit Testing:** Each function/module validated.
- **Integration Testing:** Verify seamless interaction between modules.
- **System Testing:** Validate business workflows under realistic loads.
- **User Acceptance Testing (UAT):** KEMSA verifies compliance and usability.

QA Metric	Target
Test Coverage	$\geq 85\%$
Route Calculation Accuracy	$\geq 98\%$
System Uptime (SLA)	$\geq 99.5\%$
Response Time	≤ 3 s per route request
Mean Time to Fix (MTTF)	≤ 48 hours

QA FUNNEL



8. Risk, Change, and Communication Management

Risks are proactively tracked through an Agile risk register reviewed each sprint.

Risk	Impact	Mitigation
Delayed stakeholder feedback	Medium	Weekly sprint reviews and rapid approval loops
Hardware incompatibility	High	Early environment setup and validation
Data inconsistency	Medium	Implement validation scripts and pre-processing
Scope creep	High	Backlog management and change control board
Security breach	High	Enforce encryption, access control, regular audits



9. Deployment, Training, and Handover

The deployment strategy adheres to the RFQ's specification of a high-availability Ubuntu server environment.

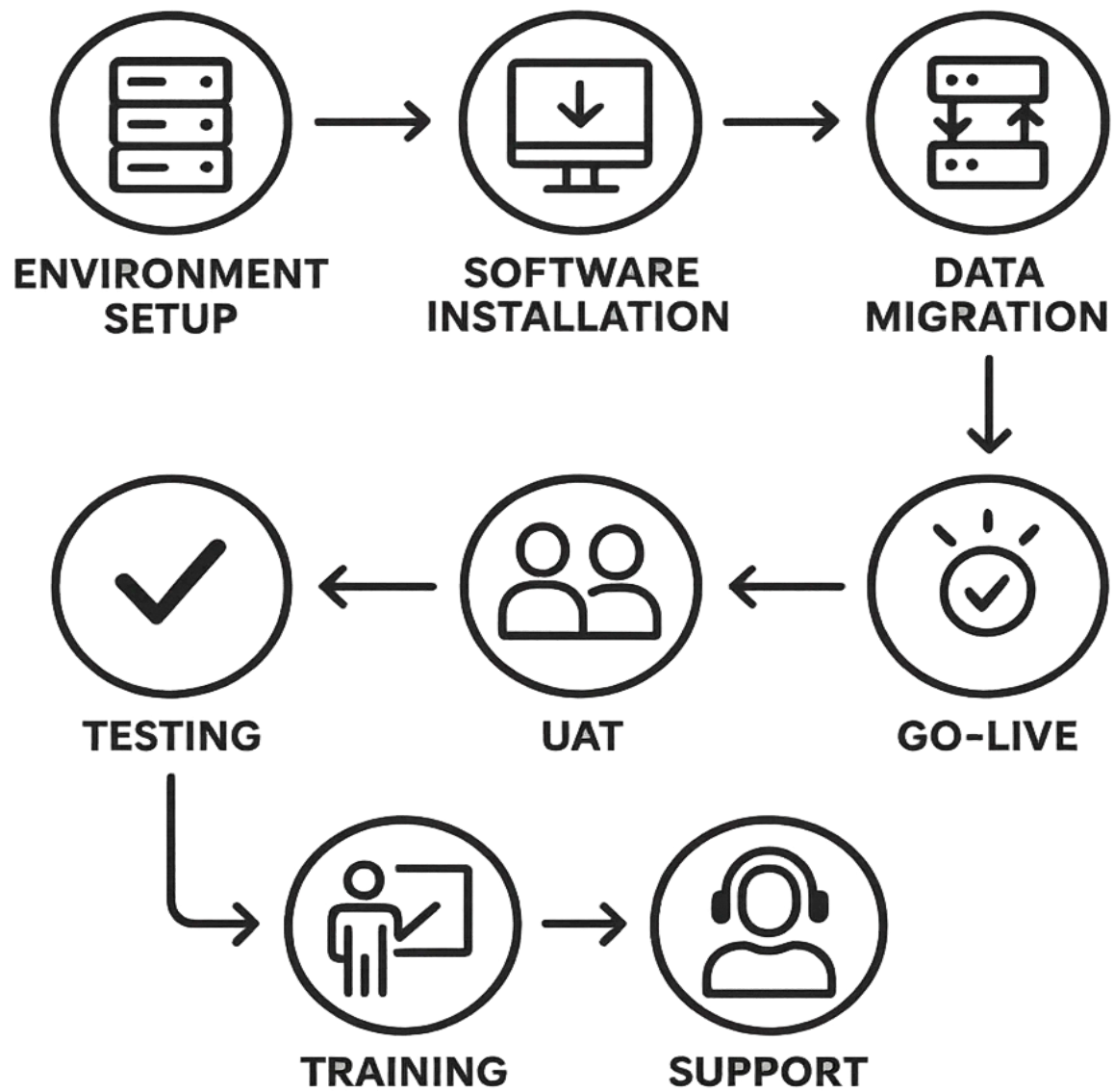
Deployment Steps

1. Prepare environment (configure RAID storage, install Ubuntu 24.04, secure network).
2. Deploy backend and database using Docker containers.
3. Install web dashboard and map visualization layer.
4. Run system verification tests and performance benchmarks.
5. Conduct UAT with the KEMSA technical team.
6. Transition to production and handover with training.

Training Plan

- **Audience:** KEMSA logistics officers, IT staff, regional managers.
- **Method:** Hands-on workshops, user manuals, video guides.
- **Outcome:** Competent use of optimization, reporting, and scheduling features.

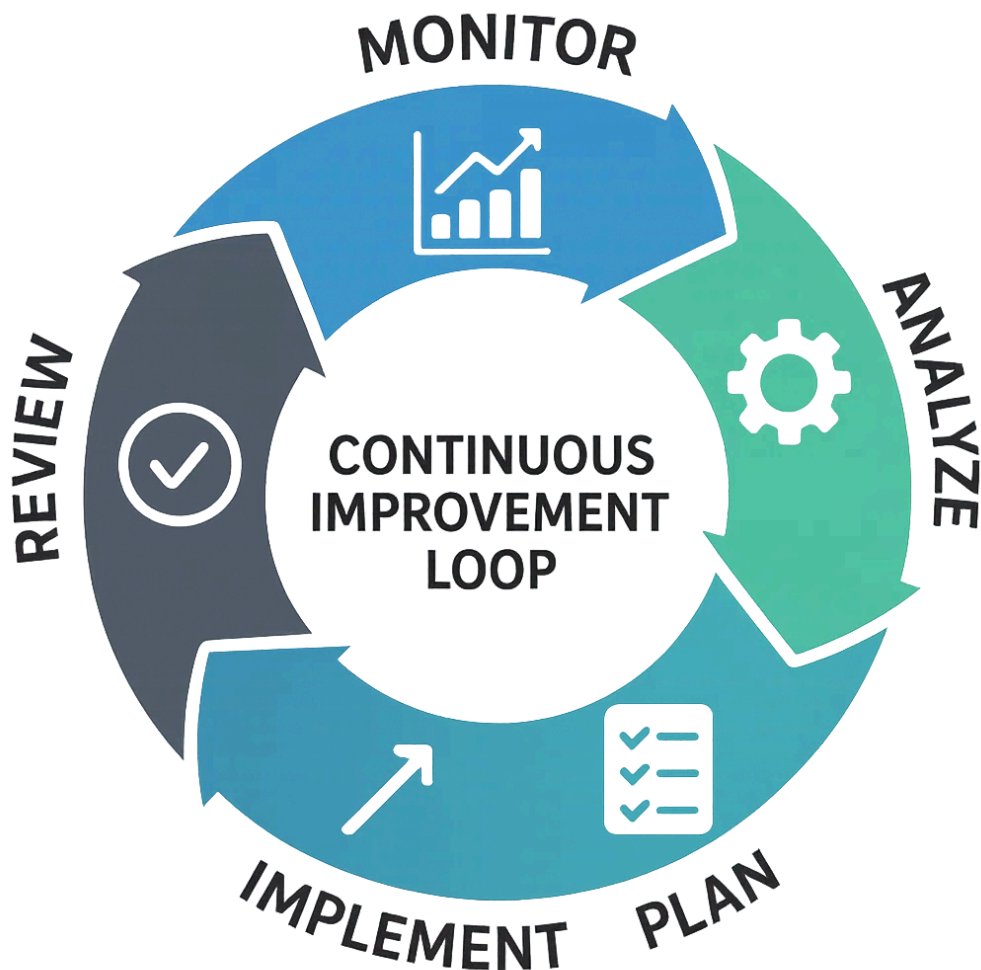
Figure 4: Deployment & Training Workflow Diagram.



10. Post-Deployment Support & Continuous Improvement

Masterspace provides a 12-month post-deployment SLA covering:

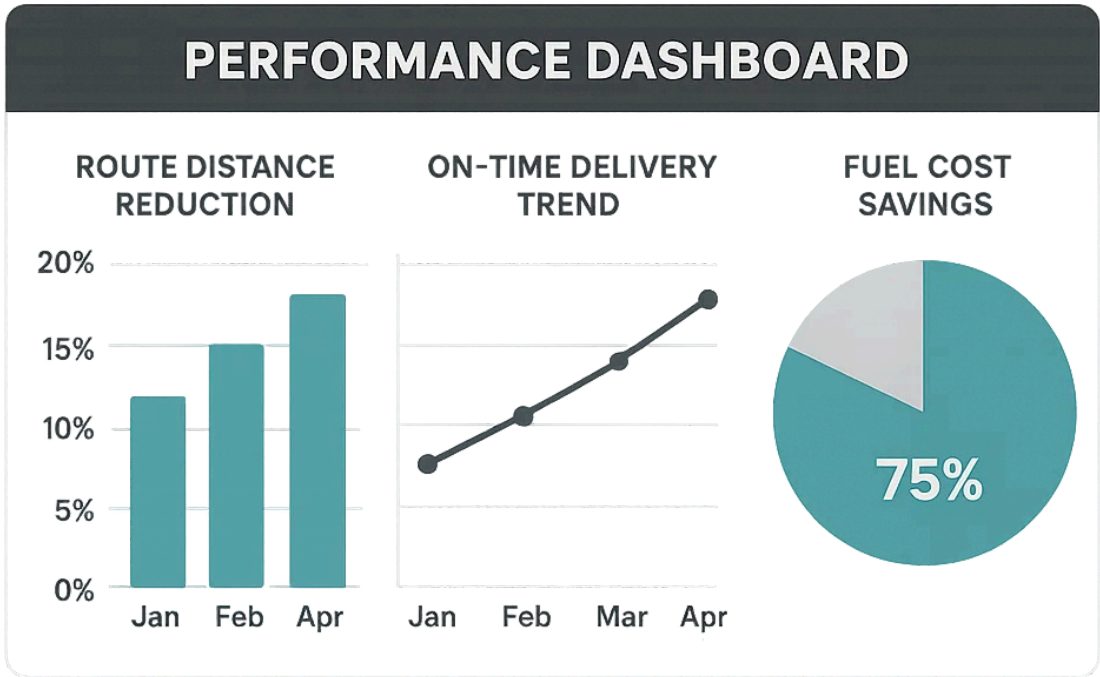
- **Level 1 Support:** User assistance and ticket management.
- **Level 2 Support:** Bug fixes, patch releases.
- **Level 3 Support:** Enhancement requests and analytics improvements.



11. Key Performance Indicators (KPIs)

KPI	Baseline	Target (Post-Implementation)
Average Route Distance	100 km per trip	≤ 80 km per trip
Average Delivery Time	8 hours	≤ 6 hours
On-Time Delivery Rate	82%	≥ 95%
Fuel Cost per Delivery	1.0 (baseline)	0.85 (−15%)
System Downtime	2 days/quarter	< 4 hours/quarter

Figure 5: Example Dashboard Visualizing Route Efficiency Improvements.



12. Conclusion

This proposed methodology integrates Agile delivery, rigorous quality control, and collaborative engagement with KEMSA. Through iterative development, early deployment, and responsive adaptation, Masterspace Solutions Ltd will deliver a robust, scalable, and user-friendly Route/Network Optimization System that aligns with KEMSA's operational goals and technical RFQ requirements.